Low-Power Wireless Uplink Utilizing Harmonic With an Integrated Rectifier–Transmitter

Xueman Sun, Changjun Liu, *Senior Member, IEEE*, Yi-Dan Chen, Jianwei Jing, Zhongqi He, *Graduate Student Member, IEEE*, and Pengde Wu, *Member, IEEE*

***Abstract*— A novel integrated rectifier–transmitter (IRT) that targets to reduce the power consumption of the uplink communication of a wireless sensor is presented in this letter. The proposed IRT fully utilizes the second harmonic generated by the rectifier as the uplink carrier and modulates this carrier in amplitude through a reconfigurable bandstop filter (BSF) that is dynamically biased by the uplink baseband signal. As a proof of concept, an IRT prototype operating at 2.4 GHz has been implemented and characterized. We report the power conversion efficiency (PCE) to prove that the modulation does not affect the rectification of IRT. An uplink data rate of 100 kb/s has been demonstrated when the IRT is fed with −20-dBm RF power.**

***Index Terms*— Rectifier, transmitter architecture, uplink, wireless power transmission (WPT).**

I. Introduction

WIRELESS sensing, as one of the main applications under the Internet of Things (IoT), faces an obvious dilemma due to the limited battery life. The battery replacements with billions of wireless sensors will lead to high environmental and maintenance costs [1], [2]. Decreasing the power consumption of sensors' transmitter and receiver is a valid solution to extend the battery life and even make battery-free devices possible.

Backscattering uplink has been the prevailing method for tag-to-reader communication in the radio-frequency identification (RFID) system [3], [4]. Although the backscattering method enables batteryless tag [5], [6], self-jamming is a major problem in backscattering radio system with the same wireless power transmission (WPT) and uplink frequency [7]–[9]. In addition to the backscattering method, the harmonic radar and the transponder-based sensing system are attractive candidates featuring high reliability, low cost, and ultralow-power consumption [10]–[13]. Researchers have already explored harmonic sensing systems, such as temperature sensor [14], IoT [15], [16], and antenna alignment [17], [18]. However, most of these uplink communications are based on the received-signal-strength indicator (RSSI) in the harmonic band [14]–[16], [18]. How to achieve digital uplink communication based on harmonic transponder still needs to explore.

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Xueman Sun, Changjun Liu, Jianwei Jing, Zhongqi He, and Pengde Wu are with the School of Electronics and Information Engineering, Sichuan University, Chengdu 610064, China (e-mail: pengdewu2018@gmail.com).

Yi-Dan Chen is with the Pillar of Engineering Product Development, Singapore University of Technology and Design, Singapore 487372.

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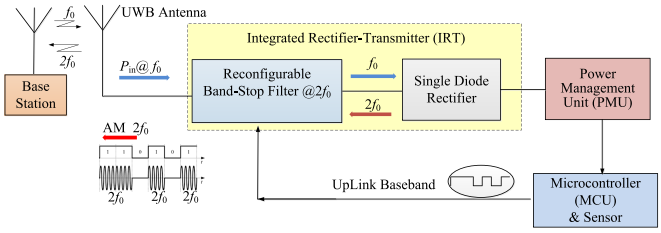


Fig. 1. System illustration of the proposed IRT, where the generated second harmonic ($2f_0$) is modulated and reradiates out through the antenna.

Generally, a transceiver for information transfer should be included in the node's schematic; however, such a transceiver requires additional power consumption in the mixing stage (i.e., dc supply to the mixer and local oscillator). To reduce the power consumption of information receiving, one line of works [19]–[21] has proposed an integrated rectifier–receiver that combines the energy harvesting and information receiving functionalities. For the uplink communication where data flow to the base station, relevant works utilizing the nonlinearity of the rectifier need to be explored.

In this letter, for the uplink communication where the data flow from the sensor to the base station, as shown in Fig. 1, we proposed a novel integrated rectifier–transmitter (IRT) in which no local oscillator/mixer is used. The second harmonic $2f_0$ generated by the rectifier is used as the uplink carrier, and a reconfigurable bandstop filter (BSF) using defected ground structure (DGS) is employed to modulate the $2f_0$ power returned to the antenna. The recovery of uplink baseband is proven possible by measurement, and the effect of the $2f_0$ modulation on the operation of the rectifier can be neglected.

II. Theoretical Analysis

As indicated in Fig. 1, the rectifier of IRT receives RF signal f_0 from the base station and converts it to dc power. Meanwhile, for a single-diode rectifier without any harmonic termination, the generated second harmonic $2f_0$ will pass through the BSF reversely and return to the antenna. To achieve amplitude modulation of the reradiated $2f_0$, a reconfigurable BSF using RF p-i-n diode and DGS is adopted in this design.

As shown in Fig. 1, the received f_0 power is denoted as P_{in}, and the insertion losses of BSF at f_0 and $2f_0$ are denoted as IL_{f_0} and IL_{2f_0}, respectively. Thus, the $2f_0$ power observed at the input side of IRT can be expressed as

$$P_{2f_0} = C_g \times (P_{\text{in}} - \text{IL}_{f_0}) - \text{IL}_{2f_0} \tag{1}$$

where C_g is the rectifier's conversion efficiency from f_0 to $2f_0$. In this design, IL_{2f0} with ON-/OFF-states of p-i-n diode can

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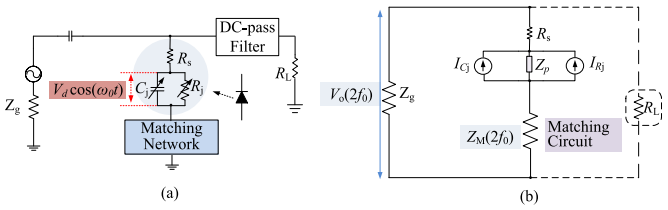


Fig. 2. (a) Block diagram of the single-diode rectifier used in IRT. (b) Diode servers as a second-harmonic source modeled by a Norton equivalent circuit.

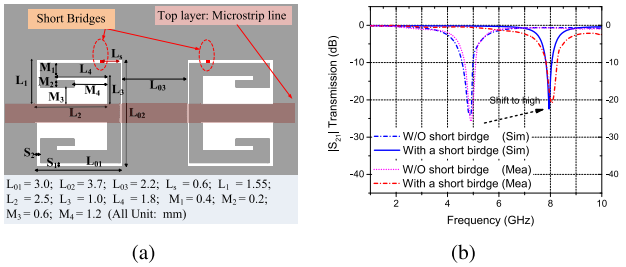


Fig. 3. BSF using DGS. (a) Physical dimension of the DGS with/without short birdies. (b) $|S_{21}|$ of the BSF with/without metal short birdies.

be described as

$$\mathrm{IL}_{2f_0} = \begin{cases} 0, & \text{Fully ON} \\ K_0 & \text{OFF} \end{cases} \tag{2}$$

in which insertion loss K_0 is determined by the Q-factor of DGS resonators at $2f_0$.

Fig. 2(a) shows the single-diode rectifier used in IRT, in which the diode junction is driven by a cosine wave that has a magnitude of V_d, and the dc-pass filter reflects all the harmonic. Fig. 2(b) shows the equivalent circuit of the rectifier in Fig. 2(a) at the second-harmonic frequency $2f_0$. According to the explicit model derived in [15], the second-harmonic currents, I_{Rj} and I_{Cj}, have a phase difference of 90°, and they are determined by the voltage across the diode junction V_d and the inherent nature of junction. Thus, at given V_d, the voltage of $2f_0$ observed at the input side with input impedance Z_g can be written as

$$V_o(2f_0) = Z_g \times |I_{Rj} + I_{Cj}| \times \frac{Z_p}{Z_g + Z_p + R_s + Z_M(2f_0)} \tag{3}$$

where Z_p is source-internal impedance of diode at $2f_0$. R_s is the junction resistance, and $Z_M(2f_0)$ represents the input impedance of the matching circuit at $2f_0$.

Based on (3), one knows that, to increase the conversion efficiency C_g, a lower $Z_M(2f_0)$ is desired. However, at the fundamental frequency f_0, the matching circuit also must compensate for the capacitive impedance of the diode

$$\mathrm{Im}(Z_D + Z_M(f_0)) = 0 \tag{4}$$

where Z_D is the diode impedance.

III. Circuit Design

A. Reconfigurable BSF

As shown in Fig. 3(a), the proposed reconfigurable BSF employs two symmetrical coupled resonators etched in the ground plane [22]. The reconfigurability of BSF is achieved by placing short bridges at the positions indicated in Fig. 3(a). Fig. 3(b) shows the simulated $|S_{21}|$ transmission of the BSF with/without short bridges. It is observed that the center

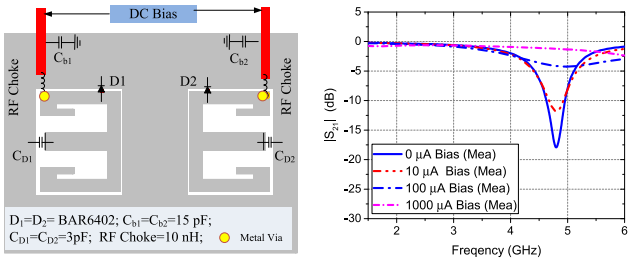


Fig. 4. (a) Bottom view of the reconfigurable BSF with p-i-n diodes and bias circuits. (b) Measured $|S_{21}|$ with bias current of 0, 10, 100, and 1000 μA.

TABLE I

$|S_{21}|$ Transmission and Bias-Power Consumption of BSF

Bias current	0 μA	10 μA	100 μA	1000 μA
Bias power of 100 kbps binary	0 μW	2.35 μW	37 μW	1.28 mW
$\lvert S_{21}\rvert$@2.4 GHz	-0.35 dB	-0.36 dB	-0.49 dB	-0.59 dB
$\lvert S_{21}\rvert$@4.7 GHz	-13.4 dB	-10.5 dB	-4.0 dB	-1.1 dB
$\lvert S_{21}\rvert$@4.8 GHz	-17.8 dB	-11.8 dB	-4.1 dB	-1.2 dB
$\lvert S_{21}\rvert$@4.9 GHz	-13.4 dB	-10.6 dB	-4.2 dB	-1.2 dB

frequency of BSF without any short bridge is 4.8 GHz, and it increases to 7.95 GHz when the short bridges are placed (blue solid line).

For experimental validation, Rogers 4350B (ϵ_r =3.66 and $\tan\delta$ =0.002) was used as the substrate. As shown in Fig. 3(b), the $|S_{21}|$ transmission has a flat passband response from 1.0 to 3.2 GHz, and the rejection band centered at 4.8 GHz is eliminated when two short bridges are placed. The simulation results were confirmed by the experimental data.

As shown in Fig. 4(a), the tunability of BSF using DGS resonators can be practically realized by replacing those short bridges with p-i-n diodes. To provide dc biasing to p-i-n diodes, the direct connections between DGS and ground plane are cutoff by slots, whereas the ac paths of DGS are recovered by two chip capacitors. Fig. 4(b) shows the measured transmission of BSF at different bias currents. It is observed that the rejection band around 4.8 GHz gradually loses its effect with the increase in the bias current. Moreover, dc power consumption and $|S_{21}|$ transmission of BSF at different bias currents are tabulated in Table I. It is observed that the reconfigurable BSF achieves the best dynamic attenuation range at 4.8 GHz.

B. Rectifier Design

As shown in Fig. 5(a), by making a tradeoff between (3) and (4), a short-circuited stub TL_1 that has a characteristic impedance of Z_1 and an electrical length of $\lambda_g/6$ at f_0 is used for diode matching. A lumped inductor L_M is applied to tune the center frequency of the rectifier during circuit debugging. The microstripe layout of the implemented rectifier is shown in Fig. 5(b) with circuit parameters.

IV. Implementation and Experimental Results

The proposed IRT was fabricated and tested. Rogers 4350B (ϵ_r =3.66, $\tan\delta$ =0.002, and h = 0.762 mm) was used as the substrate. Fig. 6 shows a photograph of the fabricated IRT, in which the rectifier has been integrated with reconfigurable

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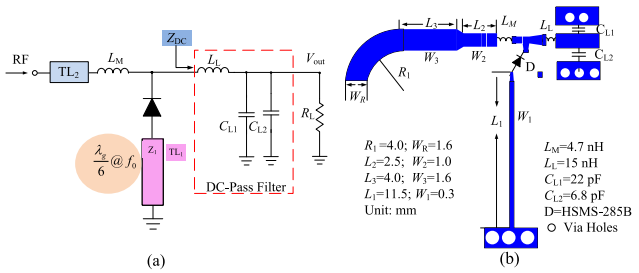


Fig. 5. Single-diode rectifier used in IRT. (a) Schematic. (b) Layout with design parameters.

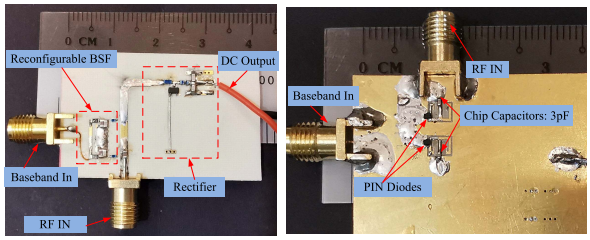


Fig. 6. Photograph of the fabricated IRT. (a) Top view. (b) Bottom view.

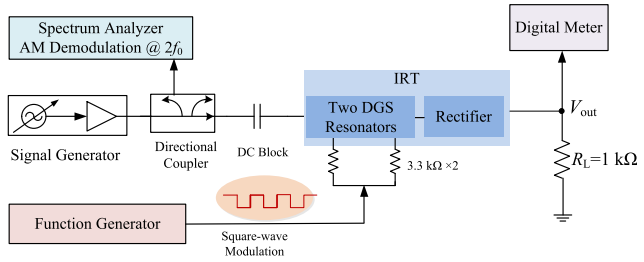


Fig. 7. Schematic of the experimental measurement setup.

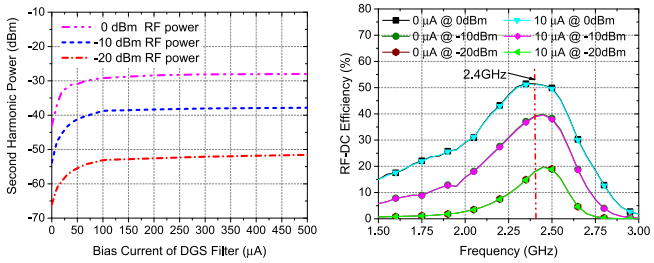


Fig. 8. (a) Measured second-harmonic level versus bias current. (b) Measured PCE versus RF frequency with/without 10-μA bias current.

BSF. Fig. 7 shows the setup of the measurement. The IRT was fed by an RF signal generator, and the second harmonic was coupled to a spectrum analyzer with the AM demodulation function. The function generator provided the uplink baseband that is a square wave of 50 kHz. For the current driving of p-i-n diodes, two 3.3 kΩ were inserted between the function generator and the bias of BSF. The dc load is $R_L = 1$ kΩ in all following experimental tests.

Fig. 8(a) depicts the 4.8-GHz harmonic level in dBm versus bias current at 0-, -10-, and -20-dBm input powers. As shown, 4.8-GHz power is proportional to the bias current ranging from 0 to 50 μA. Then, this trend slows down and reaches saturation at 100 μA. Based on the results shown in Fig. 8(a), to reduce the power consumption of modulation,

TABLE II
PCE OF IRT INCLUDING BIAS-POWER CONSUMPTION (2.4 GHz)

RF Input power	0 dBm	-10 dBm	-20 dBm
PCE without bias	51.4%	39.1%	17.9%
PCE with DC 10 μA	50.9%	34.4%	-29.1%
PCE with 50 kHz 10 μA	51.2%	36.8%	-5.6%

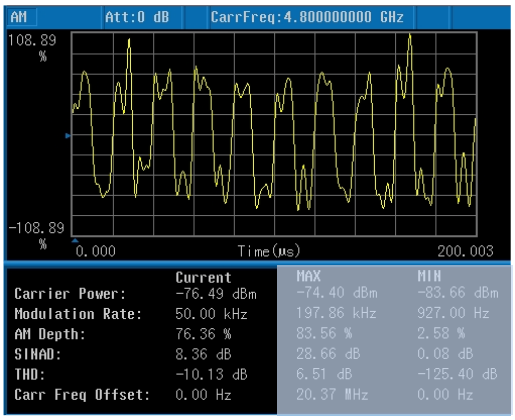


Fig. 9. Demodulated 50-kHz square-wave when the IRT has an RF input power of -20 dBm, a 20-dB coupler was used in experiments.

bias currents ranging from 0.0 to 10 μA are adapted for square-wave modulation.

To verify the effect of $2f_0$ modulation on the operation of the rectifier, we reported the power conversion efficiencies (PCEs) versus frequency at 0- and 10-μA bias currents, respectively, which are shown in Fig. 8(b). As observed at all RF power levels, the variation of the bias current causes a negligible influence on the PCE. This indicates that the operation of the rectifier at 2.4 GHz is not affected by the $2f_0$ modulation. Besides, as shown in Table II, the loss of 10-μA bias current is counted into the PCE calculation. At -20 dBm, the energy consumption of dc biasing is comparable to the harvested RF power; thus, IRT could operate at an interval mode where the harvested RF power can be accumulated and stored by the power management unit (PMU).

Finally, to test the uplink data transmission with IRT, a 50-kHz square wave (equivalent to a binary sequence) was applied to the bias of BSF, with the high-stage level of 470 mV (10-μA bias) and the low-stage level of 0 V (zero bias). When the IRT was fed with -20-dBm RF power, the modulated $2f_0$ was coupled to a spectrum analyzer (N9343C, Keysight) through a 20-dB direction coupler. The demodulated square wave is shown in Fig. 9 where the message signal of 50-kHz square wave has been well restored at the carrier power of -76.5 dBm. In another word, a 100-kb/s data rate has been achieved with an average dc power consumption of 2.35 μW.

V. CONCLUSION

In this letter, an IRT has been presented to facilitate simultaneous rectification and low-power uplink communication. The second harmonic generated by the rectifier is used as the uplink carrier and modulated by a reconfigurable BSF in amplitude. An uplink communication of 100-kb/s data rate has been demonstrated with an average dc power consumption of 2.35 μW. The reported RF-dc conversion efficiencies prove that the modulation does not affect the rectification of IRT.

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